following an interview with Willem de Sitter, a scientist who made substantial contributions to cosmology[7]). This analogy likens our three-dimensional space to the easier-to-visualize two-dimensional surface of a spherical balloon, as in Figure 8.2a, that is being blown up to larger and larger size. The galaxies are represented by numerous evenly spaced pennies glued to the balloon's surface. Notice that as the balloon expands, the pennies all move away from one another, providing a simple analogy for how expanding space drives all galaxies to separate.

An important feature of this model is that there is complete symmetry among the pennies, since the view any particular Lincoln sees is the same as the view any other Lincoln sees. To picture it, imagine shrinking yourself, lying down on a penny, and looking out in all directions across the balloon's surface (remember, in this analogy the balloon's surface represents all of space, so looking off the balloon's surface has no meaning). What will you observe? Well, you will see pennies rushing away from you in all directions as the balloon expands. And if you lie down on a different penny what will you observe? The symmetry ensures you'll see the same thing: pennies rushing away in all directions. This tangible image captures well our belief—supported by increasingly precise astronomical surveys—that an observer in any one of the universe's more than 100 billion galaxies, gazing across his or her night sky with a powerful telescope, would, on average, see an image similar to the one we see: surrounding galaxies rushing away in all directions.

And so, unlike a factory explosion within a fixed, preexisting space, if outward motion arises because space itself is stretching, there need be no special point—no special penny, no special galaxy—that is the center of the outward motion. Every point—every penny, every galaxy—is completely on a par with every other. The view from any location *seems* like the view from the center of an explosion: each Lincoln sees all other Lincolns rushing away; an observer, like us, in any galaxy sees all other galaxies rushing away. But since this is true for all locations, there is no special or unique location that is *the* center from which the outward motion is emanating.

Moreover, not only does this explanation account qualitatively for the outward motion of galaxies in a manner that is spatially homogeneous, it also explains the quantitative details found by Hubble and confirmed